

### 8.3 Analysis of the $\ell = 0$ case.

Given the simple reduction of Section 8.2, it is enough to consider the case  $\ell = 0$ . To simplify the notation, let  $C_{N,k} = C_{N,k,0}$  and  $A_{N,k} = A_{N,k,0}$ . We refer to the corresponding game as the  $(N, k)$ -growth game. Let  $\mathbb{N} = \{0, 1, \dots\}$  be the set of non-negative integers.

**Definition 8.3** (States and their cost). *Let*

$$P_{N,k} = \left\{ \mathbf{a} = (a_1, a_2, \dots, a_k) \in \mathbb{N}^k \mid \sum_{i=1}^k a_i = N \text{ and } a_i = 0 \Rightarrow a_{i-1} = 0, i = 2, \dots, k \right\},$$

be the set of all states of total size  $N$  in the  $(N, k)$ -growth game. For  $\mathbf{a} \in P_{N,k}$ , let  $C(\mathbf{a}) = C_k(\mathbf{a})$  be the minimum total cost needed to reach state  $\mathbf{a} = (a_1, a_2, \dots, a_k)$ , i.e.,  $|A_i| = a_i$ , for  $i \in [k]$ , starting from state  $(0, 0, \dots, 0)$ . Clearly  $C_{N,k} = \min\{C(\mathbf{a}) \mid \mathbf{a} \in P_{N,k}\}$ .

The following lemma gives simple recurrence relations for computing the costs of states. These relations by themselves do not provide an efficient way of computing  $C_{N,k}$  and  $C(\mathbf{a})$ , for every  $\mathbf{a} \in P_{N,k}$ . They allow us to obtain explicit formulas for these quantities, however.

**Lemma 8.4.** *For every  $\mathbf{a} = (a_1, a_2, \dots, a_k) \in P_{N,k}$ :*

- (i)  $C_k(a_1, a_2, \dots, a_k) = C_k(0, \dots, 0, a_k) + C_{k-1}(a_1, a_2, \dots, a_{k-1})$ .
- (ii)  $C_k(a_1, a_2, \dots, a_k) = \sum_{j=1}^k C_j(0, \dots, 0, a_j)$ .
- (iii)  $C_k(0, \dots, 0, a_k) = C_{a_k-1,k} + a_k$ .
- (iv)  $C(\mathbf{a}) = N + \sum_{j=i}^k C_{a_j-1,j}$ , if  $0 = a_{i-1} < a_i$ .

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*Proof.* Let  $(a_1, a_2, \dots, a_k) \in P_{N,k}$ . Consider a sequence of moves from  $(0, 0, \dots, 0)$  to  $(a_1, a_2, \dots, a_k)$ . The last move to change  $A_k$  must be to the state  $(0, \dots, 0, a_k)$ . The minimum cost of reaching this state is  $C_k(0, \dots, 0, a_k)$ . The minimum cost of moving from state  $(0, \dots, 0, a_k)$  to  $(a_1, a_2, \dots, a_k)$  is  $C_{k-1}(a_1, a_2, \dots, a_{k-1})$ , since only  $A_1, A_2, \dots, A_{k-1}$  can be used. This establishes (i); (ii) follows by induction.

The move to  $(0, \dots, 0, a_k)$  must be from a state  $(b_1, b_2, \dots, b_k)$  with  $\sum_{j=1}^k b_j = a_k - 1$ . The minimum cost of reaching such a state is exactly  $C_{a_k-1,k}$  and the cost of the move to  $(0, \dots, 0, a_k)$  is  $a_k$ . This establishes (iii); (iv) follows by induction.  $\square$

The exact formula that we obtain for  $C_{N,k}$  involves binomial coefficients. It is thus convenient to start by stating two well known binomial-coefficient identities.

**Lemma 8.5.** *For every  $n, k \geq 0$ : (i)  $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$  and (ii)  $\sum_{i=0}^k \binom{n+i-1}{i} = \binom{n+k}{k}$ .*

*Proof.* (i) is well known. (It may even be used as the definition of the binomial coefficients, along with  $\binom{0}{0} = 1$ .) (ii) is obtained by an easy induction on  $k$ . For  $k = 0$ , both terms are 1. (Note that  $\binom{n}{0} = 1$  for every  $n$ .) The induction step follows easily:

$$\sum_{i=0}^k \binom{n+i-1}{i} = \left( \sum_{i=0}^{k-1} \binom{n+i-1}{i} \right) + \binom{n+k-1}{k} = \binom{n+k-1}{k-1} + \binom{n+k-1}{k} = \binom{n+k}{k}. \quad \square$$

The following theorem gives an exact formula for computing the total cost  $C_{N,k}$  and a characterization of all the states in  $P_{N,k}$  that achieve this minimum total cost. (The corresponding optimal sequences of moves are considered in Section 8.4.)

**Theorem 8.6.** (i) If  $\binom{n+k-1}{k} - 1 \leq N \leq \binom{n+k}{k} - 1$ , for some  $n \geq 0$ , then

$$C_{N,k} = (N+1)n - \binom{n+k}{k+1}.$$

(ii) If  $\binom{n+k-1}{k} \leq N < \binom{n+k}{k}$ , for some  $n \geq 1$ , then

$$C_{N,k} - C_{N-1,k} = n.$$

(iii) If  $\binom{n+k-1}{k} - 1 \leq N < \binom{n+k}{k} - 1$ , for some  $n \geq 0$ , and  $\mathbf{a} \in P_{N,k}$ , then

$$C_{N,k} = C(\mathbf{a}) \iff \binom{n+i-2}{i} \leq a_i \leq \binom{n+i-1}{i}, \text{ for every } i \in [k].$$

Both inequalities in claim (i) are weak inequalities. Thus if  $N = \binom{n+k}{k} - 1$  for some  $n \geq 0$ , then condition (i) is satisfied by both  $n$  and  $n+1$ . This is consistent since for this value of  $N$ ,

$$\begin{aligned} (N+1)(n+1) - \binom{n+k+1}{k+1} &= (N+1)n + (N+1 - \binom{n+k+1}{k+1}) \\ &= (N+1)n + ((\binom{n+k}{k+1}) - \binom{n+k+1}{k+1}) = \underline{(N+1)n - \binom{n+k}{k+1}}, \end{aligned}$$

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where the last equality follows by Lemma 8.5(i).

Theorem 8.6(iii) can be stated more succinctly using the following definition:

**Definition 8.7** ( $Q_{N,k}$ ). If  $\binom{n+k-1}{k} \leq N < \binom{n+k}{k}$ , let

$$Q_{N,k} = \left\{ \mathbf{a} = (a_1, a_2, \dots, a_k) \in P_{N,k} \mid \binom{n+i-2}{i} \leq a_i \leq \binom{n+i-1}{i}, \text{ for every } i \in [k] \right\}.$$

Theorem 8.6(iii) says that if  $\mathbf{a} \in P_{N,k}$ , then  $C_{N,k} = C(\mathbf{a})$  if and only if  $\mathbf{a} \in Q_{N,k}$ . That is,  $Q_{N,k}$  is the set of optimal final positions in the  $(N, k)$ -growth game. For most values of  $N$  there are many optimal final positions. If  $N = \binom{n+k}{k} - 1$ , then  $((\binom{n}{1}), (\binom{n+1}{2}), \dots, (\binom{n+k-1}{k}))$  is the only optimal final state and the total cost is  $\binom{n+k}{k}n - \binom{n+k}{k-1} = \frac{kn}{k+1} \binom{n+k}{k}$ . (We say more about this in Section 8.4.)

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We prove Theorem 8.6 by induction on  $N$  and  $k$ . We say that  $(N', k') < (N, k)$  if  $N' < N$  and  $k' \leq k$ . We say that  $(N', k') \leq (N, k)$  if  $N' \leq N$  and  $k' \leq k$ . We break the proof into small pieces, each stated as a separate claim. The basis of the induction is established by the following claim:

**Claim 8.8.** Theorem 8.6 holds for  $n = 1$ , i.e., when  $0 \leq N \leq k$ .

*Proof.* If  $0 \leq N \leq k$ , then  $C_{N,k} = N$ , i.e., the insertion of each item costs exactly 1, and if  $\mathbf{a} \in P_{N,k}$ , then  $C(\mathbf{a}) = N$  if and only if  $0 \leq a_i \leq 1$ , for every  $i \in [k]$ .  $\square$

Before establishing the induction step we prove that, in a sense, Theorem 8.6(iii) implies Theorem 8.6(i), and that Theorem 8.6(i) implies Theorem 8.6(ii).

**Claim 8.9.** If Theorem 8.6(i) holds for  $(N, k)$  and  $(N-1, k)$ , then Theorem 8.6(ii) holds for  $(N, k)$ , i.e., if  $\binom{n+k-1}{k} \leq N < \binom{n+k}{k}$ , for some  $n \geq 1$ , then  $C_{N,k} - C_{N-1,k} = n$ .

(i) 可推 (ii)

*Proof.* If  $\binom{n+k-1}{k} \leq N < \binom{n+k}{k}$ , for some  $n \geq 1$ , then

$$\binom{n+k-1}{k} - 1 < N \leq \binom{n+k}{k} - 1, \quad \binom{n+k-1}{k} - 1 \leq N - 1 < \binom{n+k}{k} - 1.$$

Theorem 8.6(i) applied to  $(N, k)$  and  $(N-1, k)$  gives  $C_{N,k} - C_{N-1,k} = n$ . (Note that here we use the fact that Theorem 8.6(i) requires only the weak inequality  $N \leq \binom{n+k}{k} - 1$ .)  $\square$

**Claim 8.10.** If Theorem 8.6(i) holds for every  $(N', k') < (N, k)$ ,  $\binom{n+k-1}{k} - 1 \leq N < \binom{n+k}{k} - 1$ , for some  $n \geq 1$ , and  $\mathbf{a} \in Q_{N,k}$ , i.e.,  $\mathbf{a} \in P_{N,k}$  and  $\binom{n+i-2}{i} \leq a_i \leq \binom{n+i-1}{i}$  for every  $i \in [k]$ , then  $C(\mathbf{a}) = (N+1)n - \binom{n+k}{k+1}$ . (Note that this is the claimed value of  $C_{N,k}$ .)

*Proof.* Suppose that  $\mathbf{a} \in P_{N,k}$  and that  $\binom{n+i-2}{i} \leq a_i \leq \binom{n+i-1}{i}$ , for every  $i \in [k]$ . Then,

$$\begin{aligned} C(\mathbf{a}) &= N + \sum_{i=1}^k C_{a_i-1, i} \quad \text{8.4 (iv)} \\ &= N + \sum_{i=1}^k (a_i(n-1) - \binom{n+i-1}{i+1}) \quad \text{(i) 代入} \\ &= (N+1)n - (1 + (n-1) + \sum_{i=1}^k \binom{n+i-1}{i+1}) \\ &= (N+1)n - \sum_{j=0}^{k+1} \binom{n+j-2}{j} = (N+1)n - \binom{n+k}{k+1}, \end{aligned}$$

In the first line we used Lemma 8.4, using the fact that  $a_i \geq 1$  for every  $i \in [k]$ . In the second line we used Theorem 8.6(i) for  $(a_i - 1, i) < (N, k)$ . The third line is obtained by a simple rearrangement. The fourth line follows by Lemma 8.5. Note that

$$\sum_{j=0}^{k+1} \binom{n+j-2}{j} = \sum_{j=0}^{k+1} \binom{(n-1)+j-1}{j} = \binom{(n-1)+(k+1)}{k+1} = \binom{n+k}{k+1}. \quad \text{// 递推第k部分} \\ \text{(i)} \Rightarrow \text{(iii) 右侧值成立}$$

**Claim 8.11.** Suppose that  $\binom{n+k-1}{k} - 1 \leq N \leq \binom{n+k}{k} - 1$ , for some  $n \geq 1$ , and that  $\mathbf{a} \in P_{N,k}$ .

- (i) If  $a_i < \binom{n+i-2}{i}$  for some  $i \in [k]$ , then there exists  $j \in [k]$  such that  $a_j > \binom{n+j-2}{j}$ .
- (ii) If  $a_j > \binom{n+j-1}{j}$  for some  $j \in [k]$ , then there exists  $i \in [k]$  such that  $a_i < \binom{n+i-1}{i}$ .

*Proof.* (i) If  $a_j \leq \binom{n+j-2}{j}$  for every  $j \in [k]$ , then

$$N = \sum_{j=1}^k a_j < \sum_{j=1}^k \binom{n+j-2}{j} = \binom{n+k-1}{k} - 1,$$

contradicting the definition of  $n$ .

(ii) Similarly, if  $a_i \geq \binom{n+i-1}{i}$  for every  $i \in [k]$ , then

$$N = \sum_{i=1}^k a_i > \sum_{i=1}^k \binom{n+i-1}{i} = \binom{n+k}{k} - 1,$$

again contradicting the definition of  $n$ . // 证不可能全在下 / 全在上 □

**Claim 8.12.** If Theorem 8.6(ii) holds for every  $(N', k') < (N, k)$ ,  $\binom{n+k-1}{k} - 1 \leq N < \binom{n+k}{k} - 1$ , for some  $n \geq 1$ , and  $\mathbf{a} \in P_{N,k} \setminus Q_{N,k}$ , i.e.,  $\mathbf{a} \in P_{N,k}$  but  $\binom{n+i-2}{i} \leq a_i \leq \binom{n+i-1}{i}$  is not satisfied for some  $i \in [k]$ , then there exists  $\mathbf{b} \in P_{N,k}$  such that  $C(\mathbf{a}) > C(\mathbf{b})$ . As a consequence,  $C(\mathbf{a}) > C_{N,k}$ .

*Proof.* There are two (non-exclusive) cases:

**Case 1:** There exists  $i \in [k]$  for which  $a_i < \binom{n+i-2}{i}$ .

By Claim 8.11(i), there exists  $j \in [k]$  such that  $a_j > \binom{n+j-2}{j}$ . Let  $\mathbf{b} = (b_1, b_2, \dots, b_k) \in P_{N,k}$  be such that  $b_i = a_i + 1$ ,  $b_j = a_j - 1$ , and  $b_\ell = a_\ell$  for  $\ell \in [k] \setminus \{i, j\}$ . As Theorem 8.6(ii) is assumed to hold for  $(a_i, i)$  and  $(a_j - 1, j)$ , we have  $C_{a_i, i} - C_{a_i-1, i} \leq n-2$  while  $C_{a_j-1, i} - C_{a_j-1, j} \geq n-1$ . Thus

$$C(\mathbf{b}) - C(\mathbf{a}) = (C_{a_i, i} - C_{a_i-1, i}) - (C_{a_j-1, j} - C_{a_j-2, j}) < 0. \quad \text{a}_i < \binom{n+i-2}{i} \quad \text{a}_j > \binom{n+j-2}{j}$$

**Case 2:** There exists  $j \in [k]$  for which  $a_j > \binom{n+j-1}{j}$ .

By Claim 8.11(ii), there exists  $i \in [k]$  such that  $a_i < \binom{n+i-1}{i}$ . Again, let  $\mathbf{b} = (b_1, b_2, \dots, b_k) \in P_{N,k}$  be such that  $b_i = a_i + 1$ ,  $b_j = a_j - 1$ , and  $b_\ell = a_\ell$  for  $\ell \in [k] \setminus \{i, j\}$ . As Theorem 8.6(ii) is assumed to hold for  $(a_i, i)$  and  $(a_j - 1, j)$ , we have  $C_{a_i, i} - C_{a_i-1, i} \leq n-1$  while  $C_{a_j-1, i} - C_{a_j-1, j} \geq n$ . Thus,

$$C(\mathbf{b}) - C(\mathbf{a}) = (C_{a_i, i} - C_{a_i-1, i}) - (C_{a_j-1, j} - C_{a_j-2, j}) < 0. \quad \text{// (ii)} \Rightarrow \text{(iii)} \quad \square$$

We now have all the pieces needed to prove Theorem 8.6 by induction.

*Proof of Theorem 8.6.* The basis of the induction is established by Claim 8.8, which shows that the theorem holds for  $0 \leq N \leq k$ . We thus need to show that if the theorem holds for every  $(N', k') < (N, k)$ , for some  $N > k$ , then it also holds for  $(N, k)$ . Note that if  $N > k$  and  $N \leq \binom{n+k}{k}$ , then  $n \geq 2$ . Claim 8.10 shows that for every  $\mathbf{a} \in Q_{N,k}$  we have  $C(\mathbf{a}) = (N+1)n - \binom{n+k}{k+1}$ . Claim 8.12 shows that if  $\mathbf{a} \in P_{N,k} \setminus Q_{N,k}$  then  $C(\mathbf{a}) > C_{N,k}$ . It thus follows that  $C_{N,k} = (N+1)n - \binom{n+k}{k+1}$ , establishing Theorem 8.6(i) and (iii) for  $(N, k)$ . Claim 8.9 then establishes Theorem 8.6(ii). This completes the proof of the induction step and hence the proof of Theorem 8.6. //总结  $\square$

A convenient corollary of Theorem 8.6 is the following:

**Corollary 8.13.** (i) If  $N = \binom{n+k}{k} - 1$ , then  $C_{N,k} = \frac{kn}{k+1}(N+1)$ .  
(ii) If  $\binom{n+k-1}{k} \leq N \leq \binom{n+k}{k} - 1$ , then  $C_{N,k} \geq \frac{k}{k+1}(n-1)N$  and thus  $A_{N,k} \geq \frac{k}{k+1}(n-1) \geq \frac{1}{2}(n-1)$ .

*Proof.* To prove (i) note that if  $N = \binom{n+k}{k} - 1$  then

$$C_{N,k} = \binom{n+k}{k}n - \binom{n+k}{k+1} = \binom{n+k}{k}n - \frac{n}{k+1}\binom{n+k}{k} = \frac{kn}{k+1}\binom{n+k}{k} = \frac{kn}{k+1}(N+1). \quad \text{直接代入}$$

For each  $n \geq 1$  we prove (ii) by induction on  $N$ . If  $N = \binom{n+k-1}{k}$  then,  $C_{N,k} > C_{N-1,k} = \frac{k(n-1)}{k+1}N$ . Now, if  $\binom{n+k-1}{k} < N < \binom{n+k}{k}$ , then by Theorem 8.6(ii), 每个分块从 $\binom{n+k-1}{k}$ 递增至 $\binom{n+k}{k}-1$

$$C_{N,k} = C_{N-1,k} + n \geq \frac{k(n-1)}{k+1}(N-1) + n \geq \frac{k}{k+1}(n-1)N. \quad \square$$

## 8.4 Playing optimally - the binomial counter.

An optimal solution of the  $(N, k)$ -growth game, i.e., a sequence of moves of minimum total cost, can be easily reconstructed with the help of Theorem 8.6 and Lemma 8.4. In general such sequences are not unique. The optimal sequence is unique only if  $N = \binom{n+k}{k} - 1$  for some  $n \geq 0$ .

**Lemma 8.14.** The  $(N, k)$ -growth game has a unique optimal sequence if and only if  $N = \binom{n+k}{k} - 1$  for some  $n \geq 0$ .

*Proof.* Recall that  $Q_{N,k}$  is the set of optimal final states. We first show that  $|Q_{N,k}| = 1$  if and only if  $N = \binom{n+k}{k} - 1$  for some  $n \geq 0$ . This follows easily as  $\sum_{i=1}^k \binom{n-i+1}{i} = \binom{n+k}{k} - 1 = N$ . Thus, if  $N = \binom{n+k}{k} - 1$ , then the only optimal final state is  $a_i = \binom{n-i+1}{i}$ , for  $i \in [k]$ . If  $N$  is not of this form then  $|Q_{N,k}| > 1$  as there is some freedom in choosing the individual  $a_i$ 's. Finally, if  $N = \binom{n+k}{k} - 1$ , then by Lemma 8.4, all intermediate states are also of this form and the optimal sequence is unique. //证明 unique N为边界 子问题  $C_{a_{k-1},k}$  为  $\binom{n-k+1}{k}-1$  仍为边界  $\square$

The unique optimal sequence of moves for  $N = \binom{n+k}{k} - 1$  can be expressed using a *binomial counter* whose pseudocode is given in Figure 7. In addition to  $(a_1, a_2, \dots, a_k)$ , the sizes of the subarrays, we also keep a counter  $(b_1, b_2, \dots, b_k)$ . *Initialize*( $k$ ) sets  $a_1, a_2, \dots, a_k$  and  $b_1, b_2, \dots, b_k$  to 0. It also sets  $b_{k+1}$  to  $\infty$ . *Increment*( $k$ ) increments the counter and performs a move in the  $(N, k)$ -growth game as follows: it finds the smallest  $i$  for which  $b_i < b_{i+1}$ . (It will follow, as we shall see, that  $b_1 = b_2 = \dots = b_i < b_{i+1}$ .) It then performs the  $i$ -th move in the game, i.e.,  $a_i \leftarrow 1 + \sum_{j=1}^i a_j$  while  $a_1, a_2, \dots, a_{i-1} \leftarrow 0$ . (This corresponds to merging  $A_1, A_2, \dots, A_i$ .) It then increments the counter by letting  $b_i \leftarrow b_i + 1$  and  $b_1, b_2, \dots, b_{i-1} \leftarrow 0$ . An example of a binomial counter in action, for  $k = 4$ , is given in Figure 8. //举出 unique sequence

**Lemma 8.15.** At any time,  $a_i = \binom{b_i+i-1}{i}$  for every  $i \in [k]$ .

*Increment(k):*

*Initialize(k):*

$$\begin{cases} a_1, a_2, \dots, a_k \leftarrow 0 \\ b_1, b_2, \dots, b_k \leftarrow 0 \\ b_{k+1} \leftarrow \infty \end{cases}$$

$$\begin{cases} i \leftarrow \min\{j \in [k] \mid b_j < b_{j+1}\} \\ a_i \leftarrow 1 + \sum_{j=1}^i a_j \\ b_i \leftarrow b_i + 1 \\ a_1, a_2, \dots, a_{i-1} \leftarrow 0 \\ b_1, b_2, \dots, b_{i-1} \leftarrow 0 \end{cases}$$

Figure 7: Pseudocode of the binomial counter.

| $N$ | $a_1$ | $a_2$ | $a_3$ | $a_4$ | $b_1$ | $b_2$ | $b_3$ | $b_4$ | $N$ | $a_1$ | $a_2$ | $a_3$ | $a_4$ | $b_1$ | $b_2$ | $b_3$ | $b_4$ |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-----|-------|-------|-------|-------|-------|-------|-------|-------|
| 0   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 18  | 1     | 1     | 1     | 15    | 1     | 1     | 1     | 3     |
| 1   | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 1     | 19  | 0     | 0     | 4     | 15    | 0     | 0     | 2     | 3     |
| 2   | 0     | 0     | 1     | 1     | 0     | 0     | 1     | 1     | 20  | 0     | 1     | 4     | 15    | 0     | 1     | 2     | 3     |
| 3   | 0     | 1     | 1     | 1     | 0     | 1     | 1     | 1     | 21  | 1     | 1     | 4     | 15    | 1     | 1     | 2     | 3     |
| 4   | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 22  | 0     | 3     | 4     | 15    | 0     | 2     | 2     | 3     |
| 5   | 0     | 0     | 0     | 5     | 0     | 0     | 0     | 2     | 23  | 1     | 3     | 4     | 15    | 1     | 2     | 2     | 3     |
| 6   | 0     | 0     | 1     | 5     | 0     | 0     | 1     | 2     | 24  | 2     | 3     | 4     | 15    | 2     | 2     | 2     | 3     |
| 7   | 0     | 1     | 1     | 5     | 0     | 1     | 1     | 2     | 25  | 0     | 0     | 10    | 15    | 0     | 0     | 3     | 3     |
| 8   | 1     | 1     | 1     | 5     | 1     | 1     | 1     | 2     | 26  | 0     | 1     | 10    | 15    | 0     | 1     | 3     | 3     |
| 9   | 0     | 0     | 4     | 5     | 0     | 0     | 2     | 2     | 27  | 1     | 1     | 10    | 15    | 1     | 1     | 3     | 3     |
| 10  | 0     | 1     | 4     | 5     | 0     | 1     | 2     | 2     | 28  | 0     | 3     | 10    | 15    | 0     | 2     | 3     | 3     |
| 11  | 1     | 1     | 4     | 5     | 1     | 1     | 2     | 2     | 29  | 1     | 3     | 10    | 15    | 1     | 2     | 3     | 3     |
| 12  | 0     | 3     | 4     | 5     | 0     | 2     | 2     | 2     | 30  | 2     | 3     | 10    | 15    | 2     | 2     | 3     | 3     |
| 13  | 1     | 3     | 4     | 5     | 1     | 2     | 2     | 2     | 31  | 0     | 6     | 10    | 15    | 0     | 3     | 3     | 3     |
| 14  | 2     | 3     | 4     | 5     | 2     | 2     | 2     | 2     | 32  | 1     | 6     | 10    | 15    | 1     | 3     | 3     | 3     |
| 15  | 0     | 0     | 0     | 15    | 0     | 0     | 0     | 3     | 33  | 2     | 6     | 10    | 15    | 2     | 3     | 3     | 3     |
| 16  | 0     | 0     | 1     | 15    | 0     | 0     | 1     | 3     | 34  | 3     | 6     | 10    | 15    | 3     | 3     | 3     | 3     |
| 17  | 0     | 1     | 1     | 15    | 0     | 1     | 1     | 3     | 35  | 0     | 0     | 0     | 35    | 0     | 0     | 0     | 4     |

Figure 8: The binomial counter for  $k = 4$ .

*Proof.* By induction on the number of steps of the counter. The condition is satisfied initially, since  $\binom{i-1}{i} = 0$  for  $i \in [k]$ . Suppose that the claim holds for the current  $(a_1, a_2, \dots, a_k)$  and  $(b_1, b_2, \dots, b_k)$  and that  $b_1 = b_2 = \dots = b_i = n < b_{i+1}$ . Let  $a'_i = 1 + \sum_{j=1}^i a_j$  and  $b'_i = b_i + 1$  be the new values of  $a_i$  and  $b_i$ . By the induction hypothesis and Lemma 8.5(ii) we get

$$a'_i = 1 + \sum_{j=1}^i a_j = 1 + \sum_{j=1}^i \binom{n+i-1}{i} = \sum_{j=0}^i \binom{n+i-1}{i} = \binom{n+i}{i} = \binom{b'_i+i-1}{i}. \quad \square$$

We thus get:

**Lemma 8.16.** *If  $b_1 = b_2 = \dots = b_k = n$  for some  $n \geq 0$ , then  $N = \sum_{i=1}^k a_i = \binom{n+k-1}{k} - 1$  and the sequence of moves of the binomial counter is the unique optimal solution of the  $(N, k)$ -growth game.*

## 8.5 Extended rules.

In this section we show that the rules of the  $(N, k)$ -growth game can be slightly extended without changing the optimal values. The third rule of the game, shown in Figure 6(c) can be written as  $(a_1, a_2, \dots, a_k) \rightarrow (0, \dots, 0, 1 + \sum_{\ell=1}^i a_\ell, a_{j+1}, \dots, a_k)$ , for some  $i \in [k]$ . We first allow a move  $(a_1, a_2, \dots, a_k) \rightarrow (0, \dots, 0, a_1, \dots, a_{i-1}, \sum_{\ell=i}^j a_\ell, a_{j+1}, \dots, a_k)$ , for  $1 \leq i < j \leq k$ , in which  $A_i, A_{i+1}, \dots, A_j$  are merged into a new  $A_j$ , the subarrays  $A_1, \dots, A_{i-1}$  become  $A_{j-i+1}, \dots, A_{j-1}$  and  $A_1, \dots, A_{j-i}$  become empty. (No new item is added by this rule.) Many of the algorithms described in the paper use such moves. The cost of the move is  $\sum_{\ell=i}^j a_\ell$ . Note that the order of the items in the subarrays is still maintained.

复制使用空间, 前面移位可  
直接改指针?

**Lemma 8.17.** *Allowing moves  $(a_1, a_2, \dots, a_k) \rightarrow (0, \dots, 0, a_1, a_2, \dots, a_{i-1}, \sum_{\ell=i}^j a_\ell, a_{j+1}, \dots, a_k)$ , for  $1 \leq i < j \leq k$ , does not change the optimal values of the  $(N, k)$ -growth game.*

*Proof.* We show that for every  $\mathbf{a} \in P_{N,k}$  and every  $1 \leq i < j \leq k$  we have

$$C_k(0, \dots, 0, a_1, a_2, \dots, a_{i-1}, \sum_{\ell=i}^j a_\ell, a_{j+1}, \dots, a_k) \leq C_k(a_1, a_2, \dots, a_k) + \sum_{\ell=i}^j a_\ell,$$

showing that the new moves do not offer any advantage. By Lemma 8.4 we have  $C_k(a_1, a_2, \dots, a_k) = C_j(a_1, a_2, \dots, a_j) + C_k(0, \dots, 0, a_{j+1}, \dots, a_k)$  and similarly for the term appearing on the left hand side of the inequality. It is thus enough to prove the inequality for  $j = k$ . Now

$$\begin{aligned} & C_k(0, \dots, 0, a_1, a_2, \dots, a_{i-1}, \sum_{\ell=i}^k a_\ell) \\ & \leq C_{k-1}(0, \dots, 0, a_1, a_2, \dots, a_{i-1}) + C_k(0, \dots, 0, \sum_{\ell=i}^k a_\ell) \\ & \leq C_{k-1}(0, \dots, 0, a_1, a_2, \dots, a_{i-1}) + C_k(0, \dots, 0, a_i, a_{i+1}, \dots, a_k - 1) + \sum_{\ell=i}^k a_\ell \\ & \leq C_{k-1}(0, \dots, 0, a_1, a_2, \dots, a_{i-1}) + C_k(0, \dots, 0, a_i, a_{i+1}, \dots, a_k) + \sum_{\ell=i}^k a_\ell \\ & \leq C_k(a_1, a_2, \dots, a_k) + \sum_{\ell=i}^j a_\ell. \end{aligned}$$

*j=k 即上式用 8.4 换成  $C_j$*

The last inequality follows by expanding the terms using Lemma 8.4(iv). Note that  $C_k(a_1, a_2, \dots, a_k)$  expands to  $\sum_{i: a_i > 0} C_{a_i-1}(\dots)$ , while the previous line expands to  $\sum_{\ell: a_\ell > 0} C_{a_\ell-1}(\dots) + \sum_{\ell=1}^k a_\ell$ , where  $k_\ell \geq \ell$ , for every  $\ell \in [k]$ . More precisely, if  $1 \leq \ell \leq i-1$ , then  $k_\ell = (k-i) + \ell \geq \ell$ , and if  $i \leq \ell \leq k$ , then  $k_\ell = \ell$ . (Clearly  $C_{N,k'} \leq C_{N,k}$ , for  $k' \geq k$ .)  $\square$

*即  $a_i$  在该行的位置前添 0 所以  $k_\ell \geq \ell$  必成立*

*// 不改变最优值*

We next show that allowing general merge operations that do not necessarily respect the order of the items in the subarrays also does change the solution of the  $(N, k)$ -growth game. More precisely, for every  $I = \{i_1, i_2, \dots, i_r\} \subseteq [k]$ , where  $i_1 < i_2 < \dots < i_r$ , we allow an  $I$ -move that merges  $A_{i_1}, A_{i_2}, \dots, A_{i_r}$ , and a new item. The new subarray becomes the new  $A_{i_r}$ . The subarrays  $A_{i_1}, A_{i_2}, \dots, A_{i_{r-1}}$  become empty. All other subarrays are unchanged. In other words,  $a_{i_r} \leftarrow 1 + \sum_{\ell=1}^r a_{i_\ell}$ , while  $a_{i_\ell} \leftarrow 0$ , for  $\ell \in [r-1]$ . The standard moves in the growth games are  $[i]$ -moves, for  $i \in [k]$ . (Note that the decision to let  $A_{i_r}$  be the new merged array is arbitrary, since  $i_r$  is just the index assigned to the new array. This, however, is the convenient choice, since otherwise we need to define a new cost function  $\bar{C}(\mathbf{a})$  on states equal to the minimum of  $C(\mathbf{a}')$ , where  $\mathbf{a}'$  is a permutation of  $\mathbf{a}$ .) We can also allow a variant of an  $I$ -move in which a new item is not added. The proof of the following lemma is essentially the same as the proof of Lemma 8.17 and is therefore omitted.

**Lemma 8.18.** *Allowing  $I$ -moves for any  $I \subseteq [k]$ , does not change the optimal values of the  $(N, k)$ -growth game.*

With these extensions, the growth game now captures all the moves that can be performed by standard implementations. (See Definition 7.1.)

## 9 Lower bound on the amortized cost of grow operations

We can now prove an  $\Omega(r)$  lower bound on the amortized cost of grow operations for standard implementations that use only  $N + O(rN^{1/r})$  space to store an array of size  $N$ , showing that the constructions of Section 6 are essentially optimal.

**Theorem 9.1.** *Any standard resizable array data structure that uses only  $N + O(rN^{1/r})$  space to store an array of size  $N$ , where  $r = O(\log N)$ , must have an amortized cost of  $\Omega(r)$  for grow operations.*

*I-move 即不连续版本*



*Proof.* Assume that the extra space used by the data structure is  $rN^{1/r}$ . (If the extra space is  $\alpha rN^{1/r}$  for some constant  $\alpha > 1$ , apply the theorem for  $r - 1$ , relying on the inequality  $\alpha rN^{1/r} < (r - 1)N^{1/(r-1)}$  for sufficiently large  $N$ .) Since the operations of any standard implementation correspond to moves in the (extended)  $(N, rN^{1/r}, rN^{1/r})$ -growth game, Corollary 8.13 implies that the amortized cost of grow operations is at least  $A_{N, rN^{1/r}, rN^{1/r}} = A_{\frac{1}{r}N^{1-1/r}, rN^{1/r}} \geq \frac{1}{2}n$ , provided that  $\binom{n+rN^{1/r}}{n} < \frac{1}{r}N^{1-1/r}$ . We thus want to show that when  $n = \beta r$ , for some  $\beta > 0$ , we have  $\binom{\beta r + rN^{1/r}}{\beta r} < \frac{1}{r}N^{1-1/r}$  for sufficiently large  $N$ .

To upper bound binomial coefficients, we use the well known inequality  $\binom{n}{k} \leq \left(\frac{en}{k}\right)^k$ . We now have:

$$\binom{\beta r + rN^{1/r}}{\beta r} \leq \binom{(\beta + 1)rN^{1/r}}{\beta r} \leq \left(\frac{e(\beta + 1)N^{1/r}}{\beta}\right)^{\beta r} = \left(\frac{e(\beta + 1)}{\beta}\right)^{\beta r} N^{\beta}.$$

Choosing for example  $\beta = \frac{1}{2}$ , and assuming that  $3 \leq r \leq \alpha \log N$ , for an appropriate choice of  $\alpha$ , we get that  $r(3e)^{r/2}N^{1/2} \leq N^{3/4} \leq N^{1-1/r}$ , as required.  $\square$

## 10 Concluding remarks

We have presented a family of resizable array implementations that provide optimal trade-offs between the space needed to store an array, the space needed to resize, i.e., grow or shrink, the array, and the amortized cost of grow and shrink operations, while still maintaining  $O(1)$  worst-case access cost. We believe this solves a very fundamental problem almost completely.

Our lower bound on the amortized cost of grow operations applies only to what we call standard algorithms. We believe that these lower bounds can be extended to cover all algorithms, assuming a suitable incompressibility assumption on items and pointers. (See the discussion after the proof of Theorem 4.1.) In other words, we believe that non-standard algorithms do not offer any advantage over standard algorithms and that to prove this rigorously should not be conceptually hard.

To obtain the amortized lower bounds on grow operations, we defined an interesting growth game and then solved it exactly. While we believe that the exact solution of the game is illuminating, it would be interesting to know if there is a simpler way of obtaining, perhaps by induction, lower bounds on the values of the game that would be sufficient to prove asymptotically optimal amortized lower bounds.

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